



Biology S Review Guide

Writer: Yuna 10 (3)

Editor: Felix 10 (3)

Range: 12.1 – 12.3, 13.1 – 13.4

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

12.1 Identifying The Substance of Genes

Role of DNA

- **Storing** information (The genetic material stores information needed by every living cell (Ion transport, movement, cell growth, respiration))
- **Copying** information -> before a cell divides, its genetic information must be copied
- **Transmitting** information -> when a cell divides, each daughter cell must get a complete copy of genetic information

Experimental Proof of DNA as Genetic Material

Frederick Griffith's Experiments on Bacterial Transformation

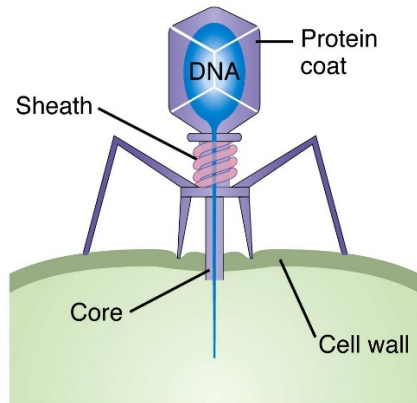
- 2 different pneumonia bacterial strains -> test of mice
 - Smooth Strain (S Strain) -> disease causing, grow into smooth colonies
 - Rough Strain (R Strain) -> harmless, grow into colonies with rough edges
- 1. Disease-causing bacteria (S Strain) -> inject in mouse -> mouse dies of pneumonia
- 2. Harmless bacteria (R Strain) -> inject in mouse -> mouse lives
- 3. Heat killed S Strain bacteria -> inject in mouse -> mouse lives
- **Conclusion:** cause of pneumonia was not a toxin from these disease-causing bacteria
- 4. Mixture of heat killed S Strain and living R Strain -> inject in mouse -> mouse dies of pneumonia -> live, disease-causing bacteria in the mouse's blood (S Strain) was found
- **Transformation:** process where one strain of bacteria is changed permanently by a gene or genes of another strain of bacteria

Oswald Avery's Experiment to Discover the Molecular Cause of Transformation

- Remove lipids, carbohydrates, protein, RNA, DNA from a solution of heat killed S Strain
- 1. Add enzymes that remove RNA (ribonuclease), lipids, carbohydrates, protein (proteinase) -> no RNA, lipids, carbohydrates, protein left in sample -> add R Strain into sample -> transformation occurs
- 2. Add enzyme to remove DNA (deoxyribonuclease) -> no DNA left in sample -> add R Strain into sample -> S Strain doesn't appear
- **Conclusion:** DNA is the molecule that stores and transmits genetic information from one generation of an organism to the next

Hershey-Chase's Experiment on Bacteriophages

- Bacteriophages



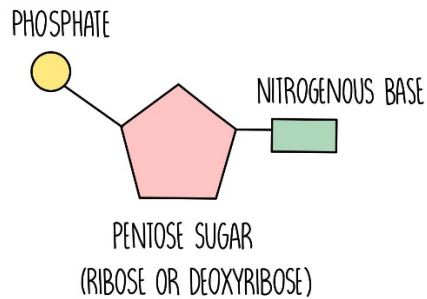
- Viruses that infect bacteria
 - Consists of protein coat and DNA/RNA core
 - How they affect bacteria -> bacteriophage latch onto bacteria -> bacteriophage DNA injected into cell, taking over bacteria's functions -> host bacteria begin producing new virus parts -> viruses are assembled, filling the bacteria -> bacteria explode and releases new viruses
- Used radioactive markers (isotopes) for experiment
 - ^{35}S (Sulfur) only mark on protein
 - ^{32}P (Phosphorus) only mark on DNA
1. ^{35}S on protein coat of virus -> bacteriophages infect bacteria with genetic material -> no radioactivity enters
 2. ^{32}P mark on DNA -> bacteriophages infect bacteria with genetic material -> radioactivity enters cell
- Conclusion: genetic material of bacteriophage was DNA not protein, DNA was the genetic material found in genes, not just in viruses and bacteria, but in all living cells -> confirmed Oswald Avery's results

12.2 Structure of DNA

Components of DNA

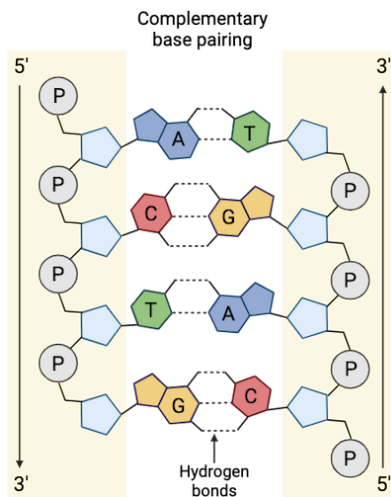
- 5C sugar (deoxyribose)
- Phosphate Group

- Nitrogen Base
 - Adenine
 - Thymine
 - Cytosine
 - Guanine



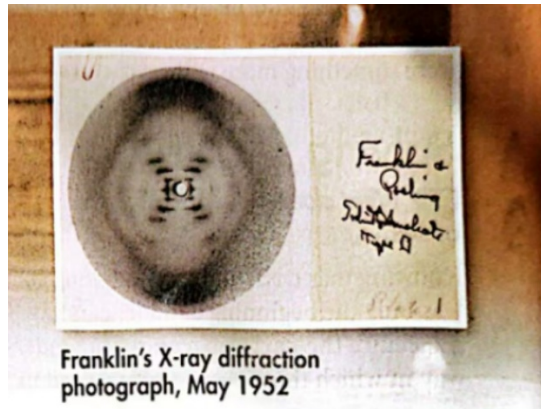
Nucleotides joined together by **covalent bonds** between 5C sugar of one nucleotide and phosphate groups of another

- **In any order**
- Deoxyribose + phosphate group is the backbone of DNA chain
- Nitrogen base in between 2 strands



Chargaff's Rule: **the amount of Adenine (A) equals the amount of Thymine (T) and the amount of Guanine (G) equals the amount of Cytosine (C)**

Franklin's X rays (Photo 51)



- Used x-rays to take photo of DNA molecules (after extracting them)
- Conclusion: DNA twisted around each other (helical structure), angle of X suggests there are 2 strands (double strand), nitrogenous base near the center

Watson and Crick's Work

- Double helix model (2 strands were wound around each other) -> with help of Franklin's X-ray

Double Helix

- 2 strands twisted -> strands are anti parallel (run in opposite directions)
- Have hydrogen bonds -> form between nitrogen base -> hold strands together
- Complementary base pairing rule -> A with T, G with C

12.3 DNA Replication

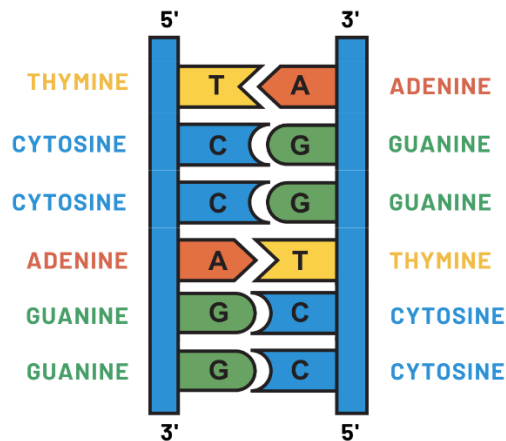
DNA replication is in S phase (interphase)

Copying -> ensures each cell will have complete set of DNA

Base pairing rules (A with T, G with C) help reconstruct sequence of new strand

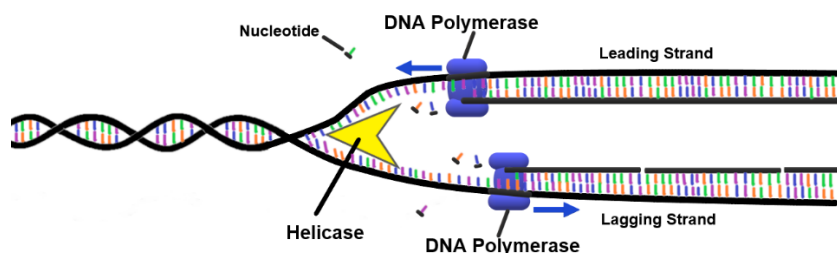
2 DNA strands are complementary -> each strand of double helix has all the information to reconstruct the other half by base pairing

Complementary DNA strand of TCCAGG



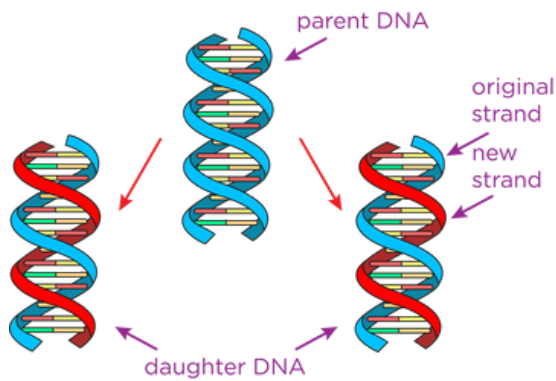
DNA replication (led by various enzymes (ends in -ase))

1. Unzip by helicase (enzyme that unwinds double helix)
 - Break hydrogen bonds between base pairs
 - DNA unwinds and 2 strands separate
2. Strands are separated, letting free nucleotides to stick on -> each strand serves as a template to produce 2 new complementary strands following base pairing rules
 - In 2 directions (by DNA polymerase)
 - Complementary bases attach to template
3. 2 identical DNA are formed (1 old strand and 1 new strand)
4. **DNA Polymerase: An enzyme that joins individual nucleotides to produce a new strand of DNA.**



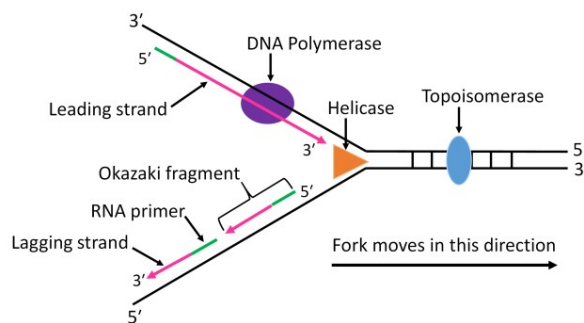
(ignore the lagging and leading strand)

DNA replication is semi-conservative replication -> at least one strand of original DNA is conserved



Enzymes: biological catalysts that speed up chemical reaction by lowering activation energy

- **Helicase**: forms **replication fork** (Y shaped region where double strands are unwound and DNA is replicated) -> unzips 2 strands of DNA by breaking hydrogen bond between base pairs



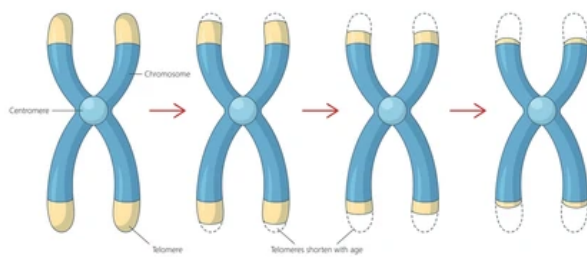
Telomere: structure of DNA

- Short, repeated DNA sequences
- **Ends** of eukaryotic chromosome
- Ends of DNA are hard to replicate
- Prevent genes from being damaged or lost during replication
- Often switched off in adults, may be activated again in cancer cell

Telomerase: enzyme

- Add short repeated DNA to telomeres to make up for loss when chromosomes are replicated
- Prevent DNA from being damaged or lost during replication

Telomere Shortening With Age



shutterstock.com · 2544246507

Prokaryotic DNA

- Located in cytoplasm
- Single, circular DNA (still have double helix)
- Usually not packed by proteins

Eukaryotic DNA

- Located in cell nucleus
- 1000 times more base pairs
- Wrap around histones
- Structure: DNA and histones -> nucleosomes: beadlike structure formed by DNA and histones
- Supercoiled chromatin
 - Chromatin: coiled DNA fibers with histones

13.1 RNA

RNA

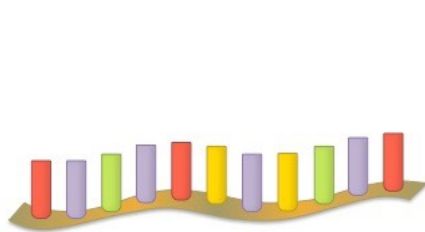
- A nucleic acid -> made up of nucleotides
- Contains decoded genetic instruction from DNA
- The information controls the assembly of amino acid into protein
- Structure
 - Phosphate Group
 - 5C Sugar -> Ribose
 - Nitrogenous Base

RNA and DNA Differences

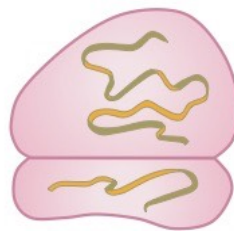
	RNA	DNA
Nitrogen Base	AUGC	ATGC
How many strands	Single strand	Double Strand
5C Sugar	Ribose	Deoxyribose
In	Nucleus	Nucleus, cytoplasm

Three Types of RNA:

1. **Messenger RNA (mRNA)** -> carry **instructions** from RNA to other parts of cell
2. **Transfer RNA (tRNA)** -> bring each amino acid to ribosome according to messages in mRNA
3. **Ribosomal RNA (rRNA)** -> ribosomes are made **up of** rRNA and proteins



Messenger RNA (mRNA)



Ribosomal RNA (rRNA)



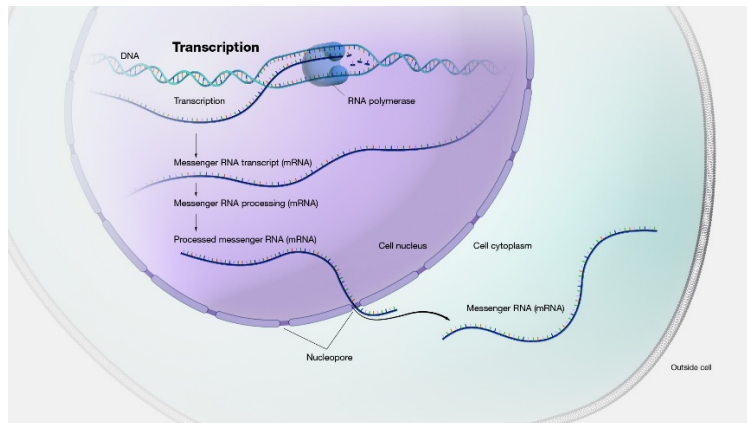
Transfer RNA (tRNA)

Transcription:

- Segments of DNA (genes) act as a template to produce complementary RNA molecules
 - A segment of DNA is copied into RNA by matching opposite bases
- **Prokaryotes** -> transcription and protein synthesis in **cytoplasm**
- **Eukaryotes** -> transcription in **nucleus**, protein synthesis in **cytoplasm**
- Requires **RNA polymerase** -> separates and builds up 2 strands
- Starts at **promoter**
 - Specific base sequences
 - Where RNA polymerase bind
 - Signals showing where to make RNA
- Ends of terminator

- Similar signals like promoter to stop transcription

1. RNA polymerase binds to promoter, unzips DNA double helix by breaking hydrogen bonds between complementary base pairs
2. RNA polymerase builds RNA by base pairing rRNA nucleotides to one strand of DNA (template)
3. RNA polymerase reaches end of gene, pre-RNA released

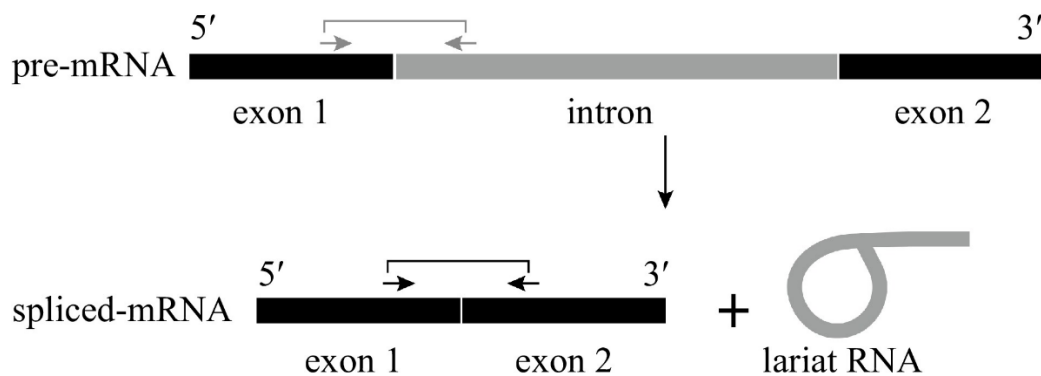


RNA polymerase

- Separates DNA molecule by **breaking hydrogen bonds** between bases
- Then **moves along** one of the DNA strands and links RNA nucleotides together

RNA **editing**: precut mRNA -> mature RNA

1. Cut out **introns** (segments of DNA that do not code for proteins)
 2. **Exons** joined back together -> splicing
 3. Cap and Tail added
- Some can combine exons in different ways -> producing different versions of mRNA (alternative splicing)



(ignore lariat RNA)

13.2 Ribosomes and Protein Synthesis

Translation

- Make polypeptide chain using genetic code on mRNA
- By ribosome in cytoplasm

Proteins

- 4 levels of organization
1. Sequence of amino acid in chain
 2. Folding/Coiling of chain
 3. Complete 3D arrangement (ionic, covalent, van der Waal forces, hydrogen bond)
 4. More than one chain

Polypeptide

- Long chain of amino acids
- Contain combination of only 20 amino acids
- Sequences of amino acid in polypeptide determine properties, shape, function of protein

Genetic Code

- Codon: 3 consecutive nucleotide bases on mRNA (triplet), codes for single amino acid
 - Start Codon: AUG – methionine
 - Stop Codon: UAA/UAG/UGA
- Some amino acids are universal -> can be coded by more than one codon

		Second letter					
		U	C	A	G		
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G	Third letter
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G	
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G	
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G	

Translation Steps

1. mRNA

- Transcribed in nucleus and enters cytoplasm
- Ribosome attaches to mRNA in cytoplasm

2. tRNA

- translation begins at **start codon (AUG)**
- tRNA brings proper amino acids into ribosome
- Ribosome attaches those amino acids to growing chain

3. Polypeptide "Assembly" Line

- Unbind bond between amino acid and its RNA
- Joins amino acid together by forming **peptide bonds**
- Ribosome **moves along**, binding new tRNA and amino acids

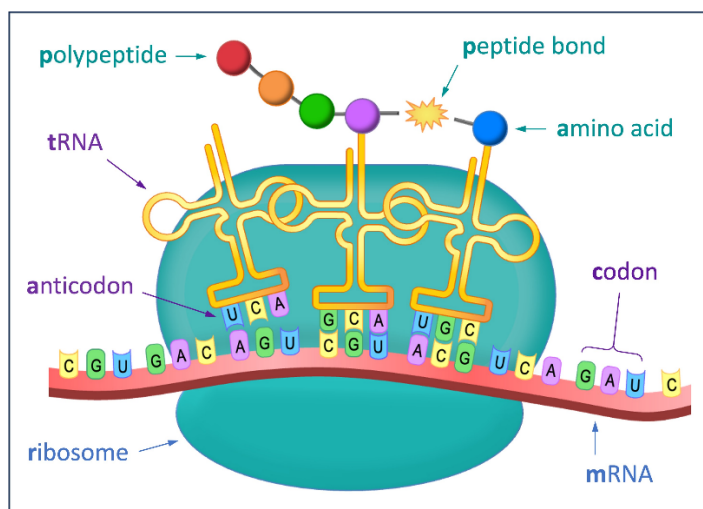
4. Completing Polypeptide

- Ribosome releases **stop codon (UAA/UGA/UAG)**
- Releases newly formed polypeptide and mRNA

- Translation ends

• Role of rRNA

- Help hold ribosomal proteins in place and help locate the beginning of mRNA message, they may even carry out the chemical reaction that joins amino acids together.



Molecular Basis of Heredity

- The **central dogma** of molecular biology
 - Genetic information is transferred from DNA to RNA to protein
- Genes contain **instruction** for making proteins
 - Enzymes catalyze and regulate chemical reactions
 - Build and operate on living cells

13.3 Mutations

Mutations: **heritable** changes in genetic information

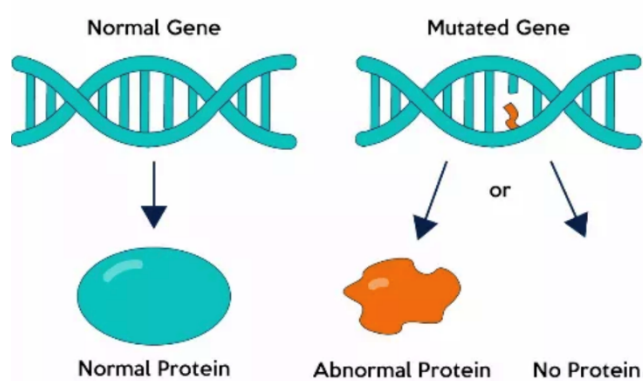
- **Mutagens**: chemical or physical agents in the environment that cause mutations
- Pesticides, tobacco smoke, pollutants, X-rays, UV light

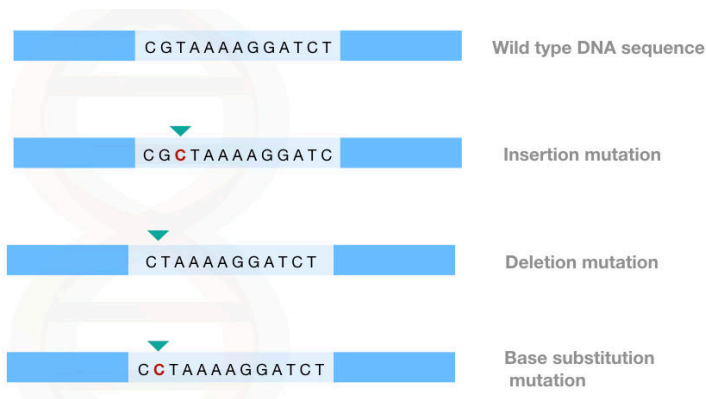
Types of Mutations

1. Gene Mutation: changes in one gene
2. Chromosomal Mutation: changes in whole chromosome

Gene Mutation

- **Point Mutation**: changes in one nucleotide
 1. Single base substitution: one nucleotide base is replaced with another
 2. Single base insertion: one nucleotide base added
 3. Single base deletion: one nucleotide is removed





Single Base Substitution

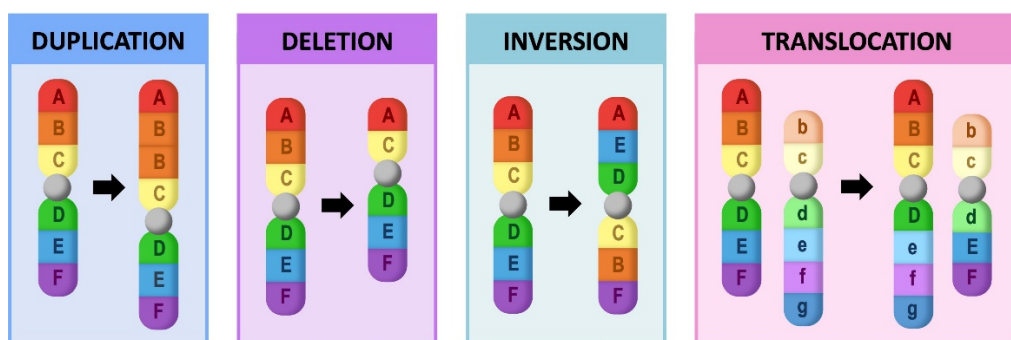
- Affect no more than one amino acid

Single Base insertion/deletion

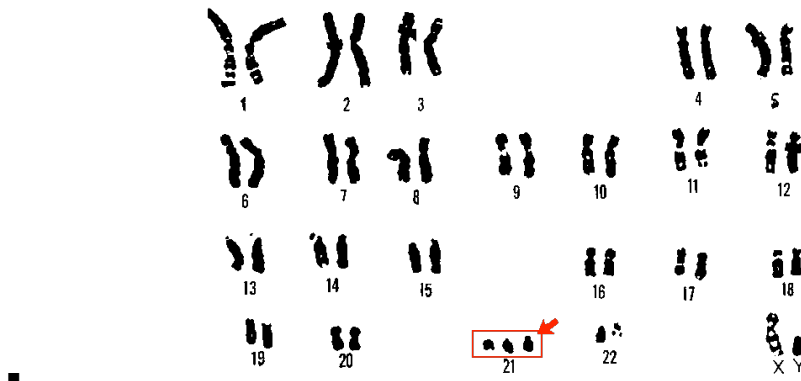
- More dramatic
- Change every amino acid following mutation point
- Altered protein -> unable to perform normal functions

Chromosomal Mutation

- Structure Change
 - Duplication: produce extra copies of parts of chromosome
 - Inversion: reverse direction of parts of chromosome
 - Translocation: part of chromosome breaks off and attaches to another
 - Deletion: loss of part or all of chromosome



- Number Change
 - **Aneuploidy:** have one less or extra chromosome ($2n+1$, $2n-1$)
 - ◆ Example: Down Syndrome -> 47 chromosomes, trisomic 21 (having 3 chromosome 21)



-
- **Polyploidy**: have three or more sets of each type of chromosome (Triploid $3n$, Tetraploid $4n$)
 - ◆ Chromosome fail to separate in meiosis
 - ◆ Polyploidy plants usually larger and stronger diploid plants

Effects of Mutations

- Many mutations are **neutral** -> have little or no effect **on gene expression** (protein function)
- **Beneficial Mutations**
 - Produce proteins with new activities useful to organisms
 - Example: insects resist pesticides, bacteria resist antibiotics
 - Source of genetic variation (for evolution)
- **Harmful Mutations**
 - Progeria – causes children to age quickly
 - Sickle Cell Disease – point mutation
 - Marfan Syndrome
 - Albinism
 - Red-Green Colorblindness
 - Amyotrophic lateral Sclerosis (ALS) – causes the nerves that control muscle to stop working

13.4 Gene Regulation and Expression

Prokaryotic Gene Regulation

- DNA binding proteins in prokaryote -> regulate genes by controlling transcription
- Organization of genes onto **operons**

- Operons: groups of genes that are regulated together
- **Example: the lac operon in *E. coli***
 - ◆ Contain 3 lactose genes
 - ◆ The three genes need to be expressed together (to produce proteins) so then the bacteria can digest lactose as food

- **Operon Structure:**

- **Promoter:** RNA polymerase binding site.
- **Operator:** Repressor protein binding site.
- **Genes:** Cluster of functionally related genes (e.g., lactose metabolism genes in *E. coli*).

1. **Eukaryotic Gene Regulation**

2. **RNA Interference (RNAi)**

Mechanism:

Small interfering RNAs (siRNAs/miRNAs) form hairpin loops → diced by **Dicer enzyme** into 20-bp fragments.

3. Fragments load into a **silencing complex** → bind complementary mRNA → block translation/degrade mRNA.

4. **Discovery (Fire & Mello, 1998):**

5. Double-stranded RNA (dsRNA) injections caused gene silencing (unc-22 gene in *C. elegans* → twitching).

6. Single-stranded RNA had no effect.

7. **Significance:**

8. Universal regulatory tool across life forms.

9. Medical potential: Silencing viral/cancer genes (RNAi technology).

10. **Cell Specialization & Differentiation**

11. **Differentiation:** Cells become specialized (e.g., liver vs. nerve cells) via selective gene expression.

12. **Challenge:** All cells share identical DNA; regulation enables diversity

Genetic Control of Development

Homeotic Genes

- **Function:** Master switches determining body part identity (e.g., fruit fly leg vs. antenna).
- **Mutation Impact:** Body part misplacement (e.g., leg growing from head).

Homeobox & Hox Genes

- **Homeobox:** 180-bp DNA sequence coding for **transcription factors** that activate developmental genes.
 - **Hox Genes:**
 - Subgroup of homeobox genes; arranged in clusters (*order correlates with body axis: head → tail*).
13. **Evolutionary Conservation:** Identical clusters in fruit flies, mice, humans (e.g., control limb/fin development).
14. **Role:** Define segment identity (e.g., Hox mutations alter organ position).